# Problem

## Title

**Automorphic Number Detector**

## Topics

Number Mathematics, Square Calculation, Digit Extraction, Loops, Conditional Statements

## Difficulty Level

Medium

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## Problem Detailed Statement (Scenario Based)

In an advanced number theory class, students are introduced to another interesting category of numbers known as **Automorphic Numbers**.

An **Automorphic Number** is a number whose square ends with the same digits as the number itself.

In simple terms, when we square the number, the last digits of the result must match the original number.

For example:

- 5
  ($5^2 = 25$) → Ends with 5 ✓
- 6
  ($6^2 = 36$) → Ends with 6 ✓
- 25
  ($25^2 = 625$) → Ends with 25 ✓
- 76
  ($76^2 = 5776$) → Ends with 76 ✓

However,

- 7
  ($7^2 = 49$) → Does not end with 7 ✗

Now, the trainer gives a positive integer **N**. Students must determine whether the given number is Automorphic or not.

To solve the problem, students must:

1. Compute the square of the number.
2. Determine the number of digits in N.
3. Compare the last digits of the square with N.
4. Use loop logic and mathematical operations (%, /).
5. Avoid converting numbers to strings.

Students must carefully handle:

- Single-digit numbers
- Large numbers
- Proper digit comparison using modulus logic

This problem strengthens:

- Mathematical manipulation
- Digit extraction
- Loop control
- Logical comparison
- Conditional structure usage

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# Input Format

A single integer:

`N`

### Constraints:

`0 ≤ N ≤ 10^6`

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# Output Format

- Print `"Automorphic Number"` if N is Automorphic.
- Otherwise print `"Not Automorphic Number"`.

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# Examples

### Example 1

#### Input:

`5`

### Output:

```
Automorphic Number
```

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## Example 2

### Input:

```
25
```

### Output:

```
Automorphic Number
```

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## Example 3

### Input:

```
7
```

### Output:

```
Not Automorphic Number
```

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## Example 4

### Input:

```
76
```

### Output:

```
Automorphic Number
```